

Excel Data Analysis ToolPak – Random Number Generation

Common setting for each example: Number of Variables = 1, Number of Random Numbers = 10, Output Range = \$A\$1.

1. Uniform Distribution

Purpose: Every value in the selected range has approximately equal chance.

Distribution: Uniform

Parameter 1 (Minimum): 1

Parameter 2 (Maximum): 10

Output Range: \$A\$1

Possible output: 3.24, 8.71, 5.12, 1.89, 9.45 ...

2. Normal Distribution

Purpose: Values follow a bell-shaped distribution around the mean.

Distribution: Normal

Mean: 50

Standard Deviation: 10

Output Range: \$A\$1

Possible output: 47.8, 62.1, 51.4, 39.7, 55.2 ...

3. Bernoulli Distribution

Purpose: Two possible outcomes, usually 0 and 1.

Distribution: Bernoulli

Probability: 0.5

Output Range: \$A\$1

Possible output: 1, 0, 1, 1, 0, 0, 1, 0, 1, 1

1 = Success/Heads; 0 = Failure/Tails.

4. Binomial Distribution

Purpose: Number of successes in a fixed number of trials.

Distribution: Binomial

Number of Trials: 10

Probability of Success: 0.5

Output Range: \$A\$1

Possible output: 6, 4, 7, 5, 3, 6, 5, 8, 4, 6

5. Poisson Distribution

Purpose: Number of events in a fixed period, based on an average rate.

Distribution: Poisson

Lambda (λ): 3

Output Range: \$A\$1

Example: A shop receives an average of 3 customers per hour.

Possible output: 2, 4, 3, 1, 5, 2, 3, 4, 2, 3

6. Discrete Distribution

Purpose: Generate only specified values according to their probabilities.

Distribution: Discrete

Value / Probability table:

1 → 0.10

2 → 0.20

3 → 0.30

4 → 0.40

Probability total = 1.00

Select the Value/Probability input range containing the two-column table.

Output Range: \$A\$1

Possible output: 4, 3, 4, 2, 4, 1, 3, 4, 2, 4

7. Patterned Distribution

Purpose: Generate numbers following a fixed pattern.

Distribution: Patterned

Start: 1

Step: 1

Output Range: \$A\$1

Output: 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

Quick Revision Table

Distribution	Parameters / Example	Meaning
Uniform	Min=1, Max=10	Equal chance in range
Normal	Mean=50, SD=10	Bell curve
Bernoulli	$p=0.5$	0 or 1
Binomial	$n=10, p=0.5$	Number of successes
Poisson	$\lambda=3$	Number of events
Patterned	Start=1, Step=1	Fixed sequence
Discrete	Values + probabilities	Specified values only

Note: Random outputs will be different each time Excel generates them. The examples shown are only sample outputs.